

Heart Failure 30-Day Readmission Prediction

Classical Binary Classification Project Report

Models: Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree

1. Project Objective

In this project, I built and compared three binary classification models to predict whether a heart-failure patient would be readmitted within 30 days. The models used were Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree. I compared them using several performance measures instead of relying only on accuracy. I also compared their training and test results to check how well they generalize and whether there were signs of overfitting.

2. Dataset and Target Variable

The dataset used for this project is `dataset_12000_records.csv`. The target column is `Readmitted_30_Days`, which tells us whether a patient was readmitted within 30 days. Around 70% of the records belong to class 0 (not readmitted), while about 30% belong to class 1 (readmitted). So, the two classes are not equally represented.

I removed the `Patient_ID` column because it is only an identifier and does not provide useful information for predicting readmission. Keeping it could also allow an identifier to affect the model even though it has no real predictive meaning.

3. Data Inspection and Preprocessing

I first checked the dataset for missing values and basic data-quality problems. There were no missing values in the working dataset. After that, I separated the input features from the target variable before training the models.

- Target: `Readmitted_30_Days`.
- Identifier removed: `Patient_ID`.
- Categorical variables were identified from object-type columns.
- Categorical columns were converted into numerical dummy variables using one-hot encoding, with the first category dropped.
- The data was divided into training and testing sets using a 70:30 split.
- I used `random_state=42` so that the same train-test split could be reproduced.
- I used stratification so that the proportion of readmitted and non-readmitted patients stayed roughly the same in both the training and test sets.

- StandardScaler was fitted only on the training data and then used to transform both the training and test data. This keeps information from the test set out of the scaling process.

Why scaling was used

Scaling puts numerical features on a similar range. This matters most for KNN because KNN uses distances to find nearby observations. If one feature has much larger numbers than the others, it can have too much influence on those distances. Scaling is also useful for Logistic Regression in this project. Decision Trees do not need scaling because they make decisions using feature thresholds rather than distances.

4. Experimental Workflow

I used the same train-test split for all three models so the comparison would be fair. For tuning the models, I used cross-validation on the training data instead of repeatedly checking the final test set.

- Logistic Regression was trained using the scaled training data.
- For KNN, I tested different values of k using five-fold cross-validation and used F1 score to choose the value.
- For the Decision Tree, I tested different maximum-depth values using five-fold cross-validation and again used F1 score for selection.
- After tuning, I evaluated the final models on the held-out test set using Accuracy, Precision, Recall, F1, and ROC-AUC.
- Confusion matrices were also examined for each model.
- I also compared training and test accuracy to get an idea of how well each model generalizes.

5. Model Descriptions and Hyperparameters

5.1 Logistic Regression

Logistic Regression was used to estimate the probability that a patient belongs to the positive class. Here, the positive class means 30-day readmission. That probability is then converted into a class prediction using a classification threshold. I used `max_iter=1000` and `random_state=42`.

5.2 K-Nearest Neighbors

KNN predicts a patient's class by looking at nearby observations. The value of k tells the model how many neighbors to consider. A small k makes the model more sensitive to individual observations, while a larger k gives a smoother result. I tested k values of 3, 5, 7, 9, 11, 15, and 21 using five-fold cross-validation and F1 score.

Selected k: 5. Best cross-validation F1 score: approximately 0.6932.

5.3 Decision Tree

A Decision Tree makes predictions by repeatedly splitting the data using feature-based rules. The maximum depth controls how many levels the tree can grow. I tested different depth values using five-fold cross-validation.

Selected maximum depth: 5. Best cross-validation F1 score: approximately 0.7065.

6. Model Performance Results

Model	Train Acc.	Test Acc.	Precision	Recall	F1	ROC-AUC	Fit diagnosis
Logistic Regression	89.64%	89.22%	84.12%	78.98%	81.47%	95.49%	Small gap; no obvious overfitting
KNN	88.87%	84.00%	82.73%	58.98%	68.86%	87.96%	Larger gap; some evidence of overfitting
Decision Tree	85.48%	84.03%	75.79%	68.70%	72.07%	88.90%	Small gap; no strong evidence of severe overfitting

From the test results, Logistic Regression had the highest accuracy, precision, recall, F1 score, and ROC-AUC in this experiment. KNN had fairly high precision but its recall was much lower. The Decision Tree had better recall and F1 than KNN, but its precision was lower.

7. Confusion Matrix Results and Healthcare Interpretation

A confusion matrix shows four types of predictions: true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP). In this project, class 1 means that the patient was readmitted within 30 days.

Model	TN	FP	FN	TP
Logistic Regression	2359	161	227	853
KNN	2387	133	443	637
Decision Tree	2283	237	338	742

In this type of healthcare prediction, a false negative means that a patient who was actually readmitted was predicted as not readmitted. A false positive means that a patient who was not readmitted was predicted as readmitted. Both types of errors matter, so the importance of precision and recall depends on how the model would actually be used.

8. Required Visualizations

I included the main visualizations needed to understand the model results: the target-class distribution, a confusion matrix for each model, a comparison of the test metrics, the combined ROC curves with AUC values, and a train-versus-test accuracy chart.

- Target distribution: approximately 70% class 0 and 30% class 1.
- Confusion matrices: included for Logistic Regression, KNN, and Decision Tree.
- Test-metric comparison: compares Accuracy, Precision, Recall, F1, and ROC-AUC.
- ROC comparison: displays all three ROC curves and their AUC values.
- Generalization chart: compares train and test accuracy for all three models.

9. ROC Curve and AUC Interpretation

The ROC curve shows how the true-positive rate changes compared with the false-positive rate as the classification threshold changes. AUC gives an overall measure of how well the model separates the two classes across different thresholds. Small fluctuations in a ROC curve are normal because the curve is based on a finite number of test observations and probability thresholds.

- Logistic Regression ROC-AUC: 0.9549
- KNN ROC-AUC: 0.8796
- Decision Tree ROC-AUC: 0.8890

10. Generalization and Overfitting/Underfitting Analysis

To check generalization, I compared the training accuracy with the test accuracy. A large drop between the two can be a sign of overfitting. If both training and test performance are low, that can point toward underfitting. However, a high training score by itself is not enough to conclude that a model is overfitting.

Model	Train Accuracy	Test Accuracy	Gap
Logistic Regression	89.64%	89.22%	0.42 percentage points
KNN	88.87%	84.00%	4.87 percentage points
Decision Tree	85.48%	84.03%	1.45 percentage points

Logistic Regression has only a 0.42 percentage-point difference between training and test accuracy, so this comparison does not show an obvious overfitting problem. KNN has the largest gap at 4.87 percentage points, which suggests weaker generalization and some evidence of overfitting. The Decision Tree has a smaller gap of 1.45 percentage points and does not show strong evidence of severe overfitting from this comparison.

11. Q1–Q14 Analysis

Q1. What is the class distribution, and why does it matter?

The dataset has about 70% class 0 and 30% class 1, so the classes are not perfectly balanced. This matters because accuracy can sometimes look good simply because the majority class is easier to predict. That is why I also looked at precision, recall, F1, ROC-AUC, and the confusion matrix.

Q2. What preprocessing was performed?

I removed Patient_ID, converted categorical columns into dummy variables, and split the data into training and test sets using stratification and random_state=42. I then fitted StandardScaler on the training data and used it to transform both datasets.

Q3. Why is scaling important, and which models need it most?

Scaling puts the features on comparable numerical ranges. It is especially important for KNN because KNN uses distances between observations. Logistic Regression can also work better with scaled features when the numerical variables have very different ranges. Decision Trees generally do not need scaling because they use threshold-based splits.

Q4. How was k selected for KNN?

I tested several k values with five-fold cross-validation on the training data. I used F1 score because I wanted to consider both precision and recall for the readmission class rather than looking only at accuracy. The selected value was k=5, with a best cross-validation F1 score of about 0.6932.

Q5. How was the Decision Tree controlled?

For the Decision Tree, I tested different maximum-depth values using five-fold cross-validation and F1 score. The selected depth was 5, with a best cross-validation F1 score of about 0.7065. Limiting the depth keeps the tree from becoming unnecessarily complex and can help reduce overfitting.

Q6. How do the models compare?

Looking at the held-out test results, Logistic Regression had the highest accuracy, precision, recall, F1, and ROC-AUC in this experiment. KNN had the lowest recall and F1. The Decision Tree performed better than KNN on recall and F1, although its precision was lower.

Q7. Which metric is especially important in a healthcare readmission problem?

Recall is important here because it tells us how many of the patients who were actually readmitted were correctly identified by the model. A missed readmission is a false negative. At the same time, precision also matters because too many false positives could lead to unnecessary follow-up or intervention. The right balance depends on how the prediction system would be used.

Q8. Is any model showing overfitting or underfitting?

KNN shows the clearest sign of a generalization issue. Its training accuracy is 88.87%, but its test accuracy drops to 84.00%, giving a 4.87 percentage-point gap. Logistic Regression has a much smaller

gap of 0.42 points, while the Decision Tree has a gap of 1.45 points. Based on accuracy alone, the results do not show clear underfitting for these models.

Q9. How could overfitting be reduced?

Overfitting can be reduced by controlling model complexity, choosing hyperparameters with cross-validation, using regularization where it makes sense, simplifying the feature set, and checking performance on data that was not used for training. For KNN, changing k can also help balance sensitivity to individual observations with excessive smoothing.

Q10. Is the model with the highest accuracy automatically the best model?

No. Accuracy is only one part of model evaluation. The classes in this dataset are not perfectly balanced, and in a healthcare setting both false negatives and false positives matter. Because of this, accuracy should be considered together with precision, recall, F1, ROC-AUC, and the confusion matrix.

Q11. What does it mean if two models have similar AUC but different precision and recall?

AUC looks at the model's ability to separate the two classes across many possible thresholds. Precision and recall, on the other hand, depend on the particular threshold used for the final predictions. This means two models can have similar AUC values but still have different numbers of false positives and false negatives at the chosen threshold.

Q12. What are the limitations of this analysis?

This analysis is based on one train-test split, so the results could change with a different sample of data. The classes are also moderately imbalanced, which means accuracy alone does not give the full picture. The three models are classical baseline models, so they may not capture every relationship in the data. Also, good predictive performance does not automatically mean that a model is ready for clinical use. Real deployment would need proper validation, calibration, clinical review, and a clear understanding of the costs of different prediction errors.

Q13. What does the model comparison suggest?

In this experiment, Logistic Regression performed strongly across the reported test metrics, and its training and test accuracies were very close. KNN had the largest train-test gap and noticeably lower recall. The Decision Tree had a smaller gap than KNN and better recall and F1, but its precision and ROC-AUC were lower than those of Logistic Regression in the reported results.

Q14. What should be considered before making a final model decision?

Before making a final model decision, I would consider how the model is going to be used, the consequences of false negatives and false positives, the evidence of generalization, all the performance metrics together, and whether the results remain stable under further validation. These results are a comparison of three baseline models in this experiment, not proof that one model will always be suitable for every heart-failure readmission dataset or clinical setting.

12. Evidence-Based Final Comparison

For this particular experiment and held-out test set, Logistic Regression gave the strongest numerical results across the reported metrics: 89.22% test accuracy, 84.12% precision, 78.98% recall, 81.47% F1, and 0.9549 ROC-AUC. Its train-test accuracy gap was only 0.42 percentage points. KNN reached 84.00% test accuracy, 82.73% precision, 58.98% recall, 68.86% F1, and 0.8796 ROC-AUC, with a 4.87-point train-test gap. Decision Tree reached 84.03% test accuracy, 75.79% precision, 68.70% recall, 72.07% F1, and 0.8890 ROC-AUC, with a 1.45-point gap.

Based on these results, I would use Logistic Regression as the main comparison model for this project. This conclusion is limited to the results from this experiment. It does not mean that Logistic Regression will always be the most suitable model for other heart-failure readmission datasets or for clinical use.

13. Limitations and Further Work

- Run repeated cross-validation or test the models on additional independent data.
- Check whether different methods for handling the class imbalance change the model comparison.
- Choose a prediction threshold based on the practical consequences of false negatives and false positives.
- Look at additional measures such as PR-AUC if the positive class becomes more imbalanced.
- Check calibration if the predicted probabilities are going to be used for risk-based decisions.
- For actual clinical use, carry out proper external and clinical validation before deploying the model.